

WERL, Germany

WMO: 104240

Lat: 51.5764N

Lon: 7.8881E

Elev: 283

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 03-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.3	21.0	9.9	9.2	17.2	15.3	12.1	23.0	30.3	47.2	26.5	46.8	4.5	60	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	85.6	66.5	81.6	65.7	78.1	64.3	68.9	80.2	67.4	77.8	65.9	75.4	8.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.5	72.6	63.7	89.1	71.0	62.2	84.5	69.5	33.2	80.2	32.0	77.9	30.8	75.4	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	19.2	16.8	9.8	93.1	8.5	3.8	3.8	95.9	-1.2	98.1	-5.9	100.2	-12.1	103.0
			9.2	71.3	8.5	1.3	3.1	72.2	-1.8	73.0	-6.6	73.7	-12.7	74.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	36.4	36.7	42.1	50.2	55.4	60.9	65.3	64.6	59.2	51.7	44.8	37.2
	DBStd	12.49	8.94	7.99	7.86	6.78	6.31	5.99	5.95	4.95	6.50	6.93	6.79	7.68
	HDD50	1809	423	375	261	80	21	1	0	0	7	63	181	397
	HDD65	5496	885	792	709	445	303	151	71	68	194	413	605	860
	CDD50	1974	3	3	16	86	189	327	474	452	283	115	25	1
	CDD65	188	0	0	0	6	27	80	54	20	1	0	0	0
	CDH74	1810	0	0	0	22	71	263	802	475	173	4	0	0
	CDH80	494	0	0	0	0	11	53	262	141	27	0	0	0
(14) Wind	WSAvg	7.7	9.8	8.8	8.5	7.0	6.8	6.4	6.6	6.7	6.8	7.1	8.6	9.1
(15) Precipitation	PrecAvg	32.9	3.0	2.5	2.4	1.9	2.4	2.8	3.4	3.3	2.8	2.7	3.0	2.9
	PrecMax	42.8	6.5	5.8	5.6	3.3	4.6	4.7	7.2	6.6	5.7	8.3	5.1	7.3
	PrecMin	24.4	0.5	0.7	0.3	0.0	1.1	0.8	0.9	1.2	0.7	0.9	0.0	1.0
	PrecStd	4.8	1.5	1.3	1.3	0.9	0.9	1.0	1.4	1.6	1.3	1.4	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	57.6	65.4	77.3	81.1	85.8	92.2	89.1	83.7	72.6	62.9	55.8
		MCWB	51.2	51.3	53.2	59.5	66.4	68.5	67.6	67.7	64.8	60.9	54.8	51.1
	2%	DB	53.4	52.9	60.9	71.9	75.4	80.7	87.3	83.8	79.3	68.3	59.5	52.0
		MCWB	50.4	47.9	51.5	58.1	61.6	66.6	66.9	65.1	64.8	60.0	53.6	48.4
	5%	DB	50.5	50.1	57.4	67.5	71.9	76.5	82.5	79.2	74.7	64.7	56.9	49.4
		MCWB	47.4	46.5	50.6	56.0	59.2	64.6	66.0	64.4	63.5	58.2	52.4	46.4
	10%	DB	47.9	47.6	53.7	63.2	68.4	73.2	78.4	74.9	70.1	62.0	54.4	47.4
		MCWB	45.1	44.7	47.8	53.6	58.3	62.9	64.9	62.9	61.4	56.7	50.8	44.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	51.9	55.4	61.0	67.8	70.7	71.4	69.9	68.4	62.9	55.9	52.0
		MCDB	54.7	56.0	61.5	72.4	78.2	81.8	81.8	84.6	76.8	69.8	60.5	54.8
	2%	WB	50.4	49.1	53.2	58.7	64.6	68.2	69.0	67.3	66.2	60.7	54.3	48.6
		MCDB	53.0	52.0	59.3	69.9	73.1	77.4	81.1	78.3	76.7	67.1	58.7	51.2
	5%	WB	47.9	46.9	50.8	56.6	61.6	66.1	67.6	65.6	64.2	58.9	52.8	46.8
		MCDB	50.5	49.7	56.2	65.6	67.9	74.2	78.9	75.4	72.4	63.8	56.8	49.4
	10%	WB	45.1	44.5	48.4	54.4	58.9	64.2	66.0	64.0	62.1	57.1	50.9	44.6
		MCDB	47.6	47.4	53.3	62.3	66.5	71.6	76.2	72.8	69.3	61.7	54.4	47.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.6	12.7	17.4	17.4	17.6	18.1	16.1	15.7	13.1	8.9	7.7
		MCDBR	8.5	11.1	19.6	25.0	24.4	23.9	26.4	24.7	23.8	16.5	11.2	8.4
	5% WB	MCWBR	7.4	7.8	11.9	13.6	13.0	12.0	9.8	9.8	11.6	9.5	7.8	7.4
		MCWBR	8.4	9.1	16.9	22.6	20.3	20.5	22.6	19.3	20.3	14.6	9.9	8.3
(40) Clear-Sky Solar Irradiance	taub		0.320	0.344	0.377	0.412	0.416	0.416	0.427	0.421	0.390	0.361	0.340	0.318
		taud	2.387	2.346	2.278	2.222	2.242	2.262	2.238	2.280	2.354	2.419	2.399	2.379
	Ebn at Noon		219	245	257	261	266	266	261	256	251	235	207	198
		Edh at Noon	19	26	34	40	41	41	41	38	31	24	19	17
(44) All-Sky Solar Radiation	RadAvg		225	438	791	1272	1539	1660	1580	1344	976	573	265	171
	RadStd		33	78	114	148	149	146	205	123	122	55	36	22

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page